

Introduction to Labour Market Economics

ECON 300 - Labour Economics

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Learning Objectives

- Understand that earnings, hours, and hourly wages are unevenly distributed.
- Learn how supply and demand determine equilibrium wages and employment.
- Identify how laws and institutions affect labour market outcomes.
- Explore different data types (time-series, cross-section, panel).
- Understand econometric tools like regression analysis.

Main Questions

- Who are the main actors in the labour market?
- How do supply and demand interact?
- What types of policy questions does labour economics address?
- What makes the labour market different from other markets?
- What is the neoclassical model of labour markets?

Main Players and Roles: Individuals

- Decide when to enter or exit the labour force.
- Choose how much education and training to pursue.
- Determine hours of work, job type, and location.
- Evaluate wages, benefits, and work-life balance.

Example: Deciding between full-time work or continuing education involves weighing opportunity costs and future wages.

Main Players and Roles: Firms

- Determine how many workers to hire.
- Decide on wage and benefit levels.
- Adjust hours, layoffs, or outsourcing.
- Plan pensions and retirement policies.

Example: A retailer may hire extra workers during holidays when expected sales rise.

Factors Influencing Firm Decisions

- **Global Competition:** Free trade, deregulation, restructuring.
- **Legislative Environment:** Minimum wage, parental leave, safety standards.
- **Changing Workforce:** Demographics, diversity, aging.

Main Players and Roles: Government

- Balances worker protection and firm competitiveness.
- Provides EI, training, pensions, and insurance.
- Shapes collective bargaining laws and standards.

Definition: Labour economics studies how wages, employment, and unemployment are determined by the interaction of labour supply and demand.

Key Idea

The labour market links people's time and skills with firms' demand for productive inputs.

Dimensions of Labour Supply

Quantity Dimensions

Population growth and participation rates.

Quality Dimensions

Education, health, and productivity.

Income Maintenance Programs

Welfare, disability, and employment insurance affect incentives to work.

Dimensions of Labour Demand

- Labour costs (wages, benefits, mandated costs).
- Demand for output (product demand drives labour demand).
- Globalization, free trade, and technological change.

- Wages, employment, unemployment, and wage differentials.
- Influenced by laws, unions, and market structure.

Sources of Income for Canadians, 2017

Table: Table 1.1: Sources of Income for Individual Canadians, 2017

	Full Sample	Men	Women
Earnings	41,348	50,824	31,910
Investment income	2,542	2,906	2,179
Pension income	1,262	1,484	1,040
Other income	952	971	933
Gov. transfers	3,728	2,495	4,955
Subtotal	49,831	58,681	41,017
Income taxes	8,391	11,168	5,625
After-tax total	41,440	47,512	35,392
Hours worked	1,411	1,593	1,230
Hourly wage	29.77	32.57	26.67

Figure 1.1: Distribution of Individual Labour Earnings (2017)

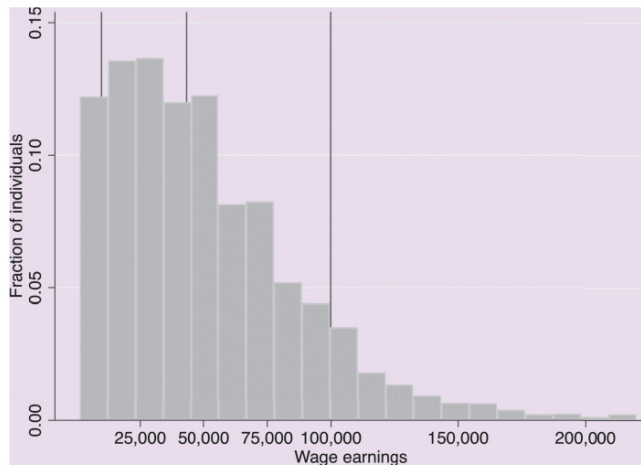


FIGURE 1.1 The Distribution of Individual Labour Earnings, 2017

Figure 1.2: Distribution of Annual Hours Worked (2017)

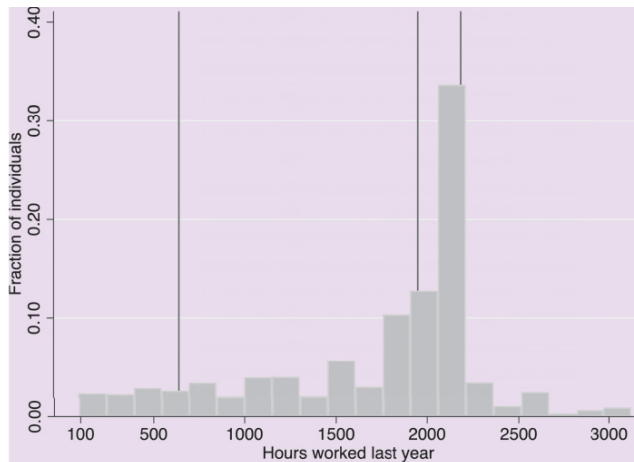


FIGURE 1.2 The Distribution of Individual Annual Hours Worked, 2017

Figure 1.3: Distribution of Average Hourly Earnings (2017)

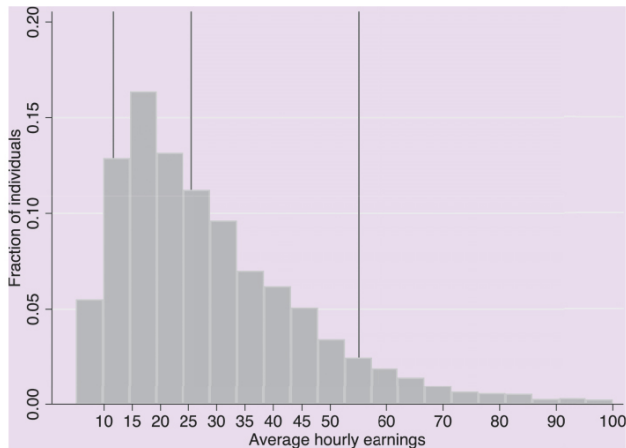


FIGURE 1.3 The Distribution of Individual Average Hourly Earnings (Wages), 2017

Neoclassical Model of Labour Supply and Demand

- Workers supply labour to maximize utility.
- Firms demand labour to maximize profit.
- Market equilibrium determines wage and employment.

Figure 1.4: Wages and Employment in a Competitive Market

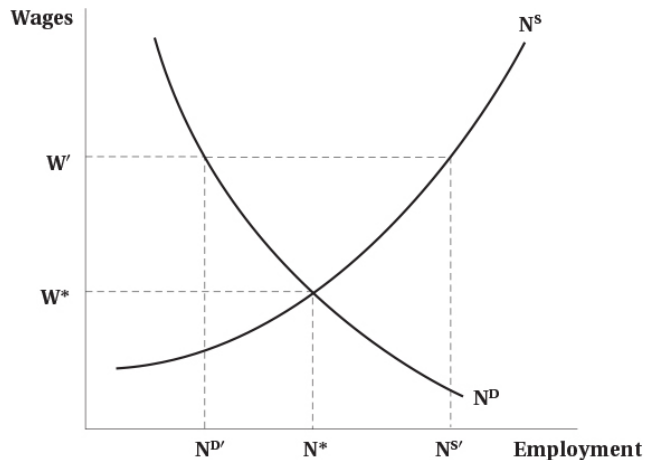


FIGURE 1.4 Wages and Employment in a Competitive Labour Market

Figure 1.5: Equilibrium Wages Across Two Sectors

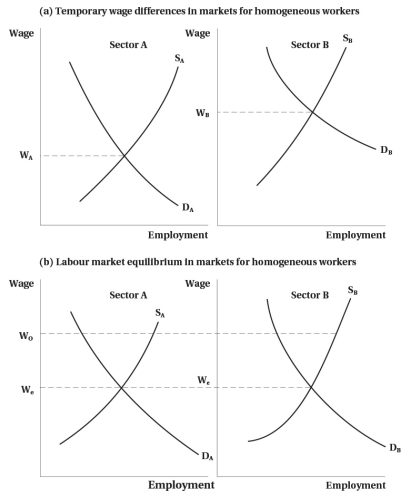


FIGURE 1.5 Equilibrium Wages across Two Sectors

- **Labour Supply:** Impact of EI, welfare, and childcare.
- **Labour Demand:** Effects of trade, outsourcing, and regulation.
- **Wage Structures:** Gender gaps, benefits, and equity.
- **Unemployment:** Youth and cyclical unemployment trends.

- Widespread layoffs and remote work adjustments.
- Disproportionate impact on low-income earners.
- Parents reduced hours due to childcare closures.

Figure 1.6: COVID-19 Employment Losses Across Earning Quartiles

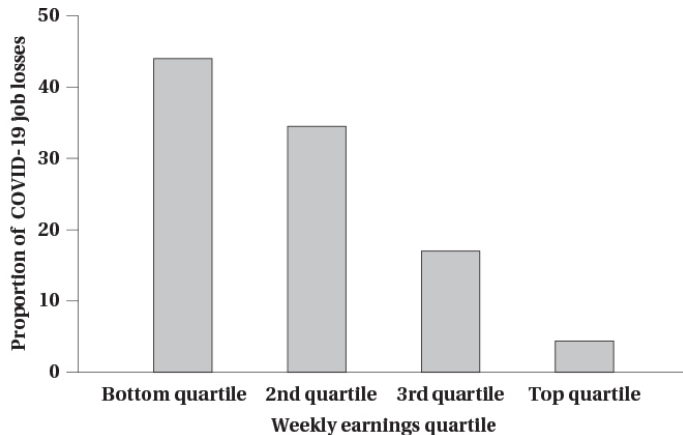


FIGURE 1.6 The Distribution of COVID-19 Employment Losses Across Earning Quartiles

Labour Market vs. Other Markets

- Actors pursue diverse goals beyond profit.
- Influenced by legal, institutional, and social factors.
- Wages serve as price, signal, and incentive simultaneously.

- Main actors: individuals, firms, and government.
- Outcomes: wages, employment, hours.
- Neoclassical model as key analytical tool.
- Institutional and policy factors modify market outcomes.

Appendix 1A: Regression Analysis and Research Design

- Data types: time-series, cross-section, and panel.
- Empirical tools link labour outcomes with determinants.
- Example: NHL player salaries and performance.

Figure 1.A1: Supply and Demand – “Star” vs. “Grinder” Players

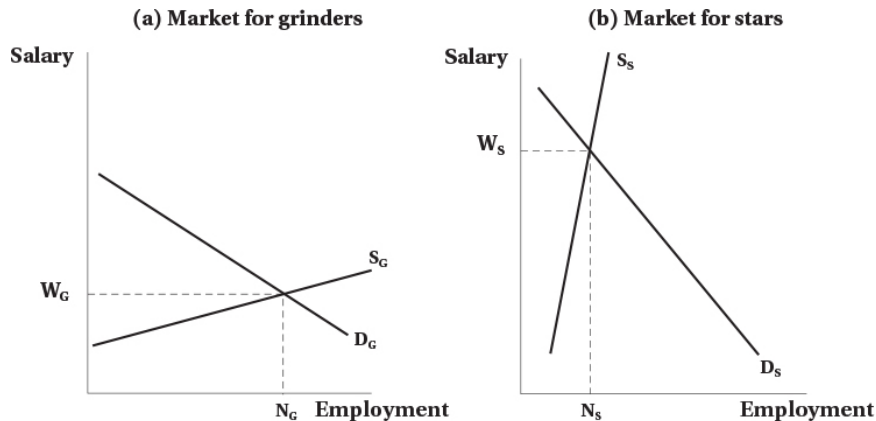


FIGURE 1.A1 Supply and Demand and the Relative Earnings of Star versus Grinder Hockey Players

Figure 1.A2: NHL Player Salaries by Goals Scored (2018–19)

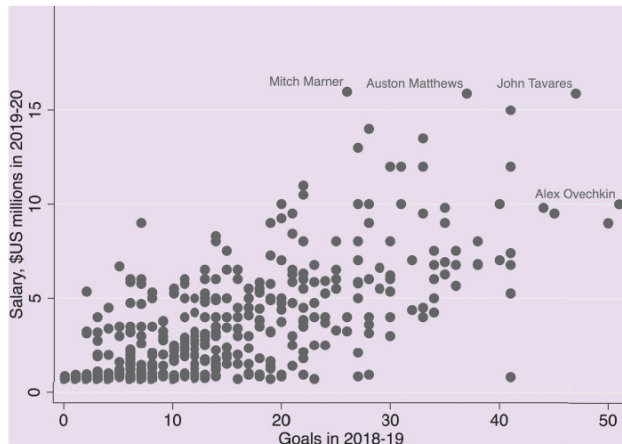


FIGURE 1.A2 NHL Player Salaries by Goals Scored in the Previous Year

Figure 1.A3: Log Salary and Goals with Regression Line (2018–19)



FIGURE 1.A3 Logarithm of NHL Player Salaries by Goals Scored in the Previous Year, and Fitted Regression Line

Tabulation of NHL Player Salaries by Goals Scored

Table: Table 1.A1: NHL Player Salaries by Goals Scored, 2018–19

Goals	Sample Size	Salary (millions)	$\ln(\text{Salary})$
0–9	121	1.93	14.20
10–19	133	3.13	14.73
20–29	76	5.53	15.32
30–39	32	7.51	15.76
40–49	11	9.03	15.82
50+	2	9.50	16.07

Estimated Effects of Performance on Salary (OLS, 2019–20)

Table: Table 1.A2: OLS Estimates of Player Salary Determinants

	(1)	(2)	(3)	(4)	(5)	(6)
Intercept	685,592	247,395	197,563	13.972	13.867	13.859
Goals	195,498	88,605	90,550	0.052	0.027	0.027
Assists		98,218	98,902		0.024	0.024
Plus/Minus			-8,156			-0.001
R^2	0.44	0.54	0.54	0.39	0.47	0.47

Appendix 1B: Research Designs in Labour Economics

- Experimental: Random assignment (treatment vs. control).
- Non-experimental: Regression, DiD, natural experiments.
- Purpose: Estimate counterfactuals—what would happen without policy change.

Thank You!